

MATTHEW MO

📞 437-982-6562 | ✉️ mzmo@uwaterloo.ca | [linkedin.com/in/matthew-mo520](https://www.linkedin.com/in/matthew-mo520) | github.com/MatthewMo520 | 🌐 matthew-mo.com

EDUCATION

University of Waterloo

Bachelor of Mathematics, Data Science (Co-op)
Cumulative Average: **85%**

Waterloo, ON

Sept. 2024 – May 2028

EXPERIENCE

Data Science Intern

Jan. 2026 – Present

Scotiabank

Toronto, ON

- Automated model performance monitoring across KPI tracking, confusion matrix generation and accuracy reporting, reducing manual Excel reporting from **2–3 hours to under 5 minutes**
- Developed a mortgage pricing simulation and benefit pipeline; implemented a configurable Pyomo optimization model with adjustable profit, volume and RoRWA constraints to identify margin-maximizing rate strategies across **100,000+** mortgage customers
- Built a customer mortgage renewal sensitivity model using K-Means clustering to segment **~20,000 monthly customers** into VH/H/M/L sensitivity tiers with ACID validation, enabling personalized rate offers targeting high-sensitivity segments to improve retention and increase profit

Machine Learning Engineer

Nov. 2025 – Jan. 2026

AgSights

Elora, ON

- Built a voice-to-database pipeline using Whisper (ASR) + spaCy for entity extraction, achieving **91% entity accuracy** on farm record inputs; reduced manual data entry time by **~60%** in field trials

Machine Learning Engineer

May 2026 – Present

Lyyvora

Remote

- Built a web scraper aggregating **3,000+** podcast outreach candidates and an LLM-powered pipeline using **Groq** to auto-generate and send personalized marketing emails, eliminating manual outreach drafting entirely

PROJECTS

Personal Finance Platform | React, FastAPI, PostgreSQL, Plaid API, scikit-learn, Docker, Railway

- Engineered a personal finance platform with real bank linking via **Plaid API**, mandatory **TOTP 2FA**, token auto-refresh, and a React dashboard with animated spending charts which is deployed on Railway with Docker.
- Trained a **Random Forest** transaction classifier on **20,000+** labeled records using **TF-IDF** feature extraction and **Isolation Forest** anomaly detection; integrated ML pipeline into **FastAPI + PostgreSQL** backend with auto-inference on new bank syncs.

LinkedIn (GeeseHacks - Best Beginner Hack) | React(JS), Tailwind(CSS), HTML, Python, Selenium

- Built a web app that automates LinkedIn job searches using Selenium web scraping, scraping 3+ pages of candidate profiles, extracting and filtering by role and company; awarded Best Beginner Hack at GeeseHacks.

Financial 10-K Q&A – RAG Pipeline | LangChain, FAISS, sentence-transformers, Groq, FastAPI, Streamlit, Docker

- Built a dynamic RAG pipeline over live SEC 10-K filings using LLM-based ticker extraction, FAISS vector search, and Llama 3 (Groq); chunked filings into **~1,000 embeddings** per company using sentence-transformers; implemented persistent cache with automatic invalidation, reducing repeated query latency from **30s to under 2s** across **6 companies**; deployed via Dockerized FastAPI + Streamlit.

FinBERT Financial Sentiment Analysis | Python, PyTorch, HuggingFace Transformers, FastAPI, Docker

- Fine-tuned **FinBERT** on **9,500+** financial news sentences for 3-class sentiment classification, achieving **89%** accuracy and **86%** macro F1; deployed via Dockerized FastAPI endpoint with confidence scoring

SKILLS

Languages: Python, R, Java, Javascript, C, C++, HTML/CSS, SQL, SAS

Libraries & Frameworks: Scikit-learn, NumPy, Pandas, PySpark, Pyomo, Matplotlib, Plotly, Streamlit, Flask, LangChain, FAISS, sentence-transformers, HuggingFace Transformers, PyTorch

Tools & Platforms: GCP, Git, Docker, Jupyter, Microsoft Dynamics, MongoDB, Excel VBA